Preface

Space and time capture the imagination like no other scientific subject. For good reason. They form the arena of reality, the very fabric of the cosmos. Our entire existence—everything we do, think, and experience — takes place in some region of space during some interval of time. Yet science is still struggling to understand what space and time actually are. Are they real physical entities or simply useful ideas? If they're real, are they fundamental, or do they emerge from more basic constituents? What does it mean for space to be empty? Does time have a beginning? Does it have an arrow, flowing inexorably from past to future, as common experience would indicate? Can we manipulate space and time? In this book, we follow three hundred years of passionate scientific investigation seeking answers, or at least glimpses of answers, to such basic but deep questions about the nature of the universe.

Our journey also brings us repeatedly to another, tightly related question, as encompassing as it is elusive: What *is* reality? We humans only have access to the internal experiences of perception and thought, so how can we be sure they truly reflect an external world? Philosophers have long recognized this problem. Filmmakers have popularized it through story lines involving artificial worlds, generated by finely tuned neurological stimulation that exist solely within the minds of their protagonists. And physicists such as myself are acutely aware that the reality we observe—matter evolving on the stage of space and time—may have little to do with the reality, if any, that's out there. Nevertheless, because observations are all we have, we take them seriously. We choose hard data and the framework of mathematics as our guides, not unrestrained imagination or unrelenting skepticism, and seek the simplest yet most wide-reaching theories capable of explaining and predicting the outcome of today's and future experiments. This severely restricts the theories we pursue. (In this book, for example, we won't find a hint that I'm floating in a tank, connected to thousands of brain-stimulating wires, making me merely *think* that I'm now writing this text.) But during the last hundred years, discoveries in physics have